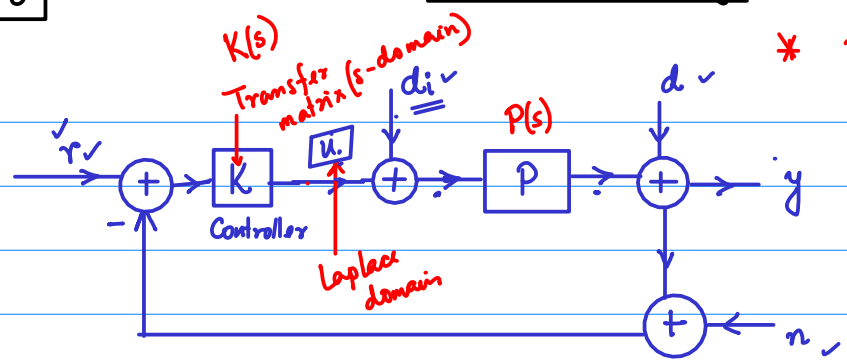
L28

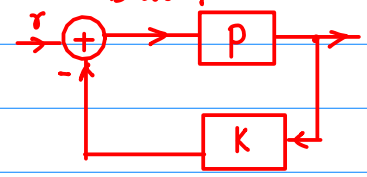
Internal Stability

03/11/2020



$r = R(s)$
$u = U(s)$
$d_i = D_i(s)$
$d = D(s)$
$y = Y(s)$
$n = N(s)$

Basic feedback structure



Control i/p : $u = K(r-n-y) = K(r-n-P(u+d_i)-d)$

$\Rightarrow (I+KP)u = K(r-n-d) - KP\,d_i$

$\Rightarrow u = (I+KP)^{-1}K(r-n-d) - (I+KP)^{-1}KP\,d_i$ $\quad$ * $(I+KP)$ is invertible assumed

O/p: $y = Pe_1 + d \Rightarrow (I+PK)\,y = PK(r-n) + Pd_i + d \Rightarrow y = (I+PK)^{-1}PK(r-n) + (I+PK)^{-1}Pd_i + (I+PK)^{-1}d$

* $(I+PK)$ is invertible.

Can we design any realizable controller and connect it to the plant?

$K(s) \leftarrow$ proper $\quad \lim_{s\to\infty} \|K(s)\| < \infty$

$\frac{s+2}{s+1}$ $\quad$ $\frac{2}{s+1}$

$\times \frac{s^2+s+2}{s+1} \times$

anticausal

e.g. $P = \frac{-(s-1)}{s+2}$ $\quad K = 1$ $\quad I+KP = \frac{3}{s+2}$

$P(\infty) = -1$
$K(\infty) = 1$
$\hat{K}(\infty) = -1$

$$\boxed{u = \left(\frac{s+2}{3}\right)(r-n-d) + \left(\frac{s-1}{3}\right)d_i}$$

Not realizable

$$\begin{bmatrix} \frac{1}{s+1} & \frac{1}{s+2} \\ 0 & \frac{s+3}{s+1} \end{bmatrix}$$ us

$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \leftarrow$ not invertible

$$K = \begin{bmatrix} 1/s+1 & 1/s+2 \\ 0 & -1 \end{bmatrix}, \quad P = \begin{bmatrix} 1/s+2 & 1/s+3 \\ 0 & 1 \end{bmatrix}$$

$\lim_{s\to\infty} \frac{-(s-1)}{s+2} = \lim_{s\to\infty} \frac{-s+1}{s+2} = -1$

$$I+KP = \begin{bmatrix} 1 + 1/(s+1)(s+2) & 1/(s+1)(s+3) + 1/s+2 \\ 0 & 0 \end{bmatrix}$$ is not invertible $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

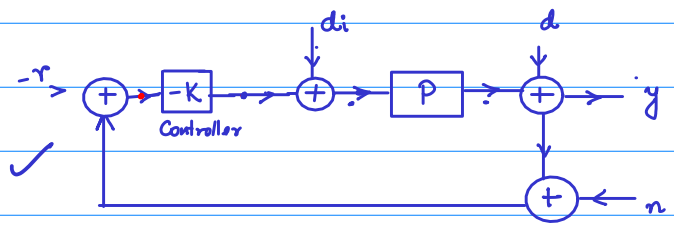
A feedback system is said to be well-posed if all closed-loop transfer matrices are well-defined and proper.

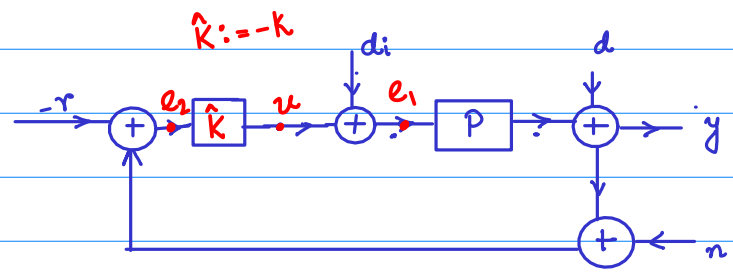
* If r, n, d & d_i are specified and u is well-defined, then y and all other signals are well-defined.

* If T.M from r, n, d & d_i to u exists are are proper, then T.M from r, n, d, d_i to any other signal (like y) exists and is proper.

* System is well-posed iff the T.M from $\begin{pmatrix} d_i \\ d \end{pmatrix} \longrightarrow u$ exists and is proper

$\boxed{r \,\&\, n} \leftarrow \begin{matrix} d \\ d_i \end{matrix}$





$$e_2 = -r + n + d + P(u + d_i) = w_2 + P e_1 \qquad w_2 := -r + n + d$$
$$e_1 = \hat{K} e_2 + d_i = w_1 + \hat{K} e_2 \qquad w_1 := d_i$$

$$\boxed{\begin{aligned} w_1 &= e_1 - \hat{K} e_2 \\ w_2 &= -P e_1 + e_2 \end{aligned}} \longrightarrow \text{I}$$

I/p to controller : e_2
I/p to the plant : e_1
External i/ps : w_1, w_2

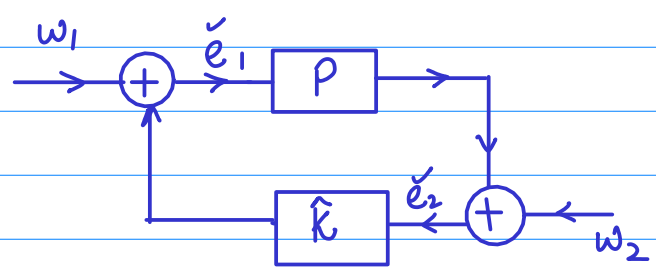


Figure 1:
Feedback structure

The system in fig (1) is well-posed if and only if

$$\begin{bmatrix} I & -\hat{K}(\infty) \\ -P(\infty) & I \end{bmatrix} \text{ is invertible}$$

$\hat{K}(\infty) = \lim_{s \to \infty} \hat{K}(s)$
$P(\infty) = \lim_{s \to \infty} P(s)$

Proof: From I:

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} I & -\hat{K} \\ -P & I \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} \Rightarrow \begin{bmatrix} w_2 \\ w_1 \end{bmatrix} = \begin{bmatrix} I & -P \\ -\hat{K} & I \end{bmatrix} \begin{bmatrix} e_2 \\ e_1 \end{bmatrix}$$

$\lim_{s\to\infty} G(s) = D$

$G(s) = D + \boxed{C(sI-A)^{-1}B}$

$G^{-1}(s)$

Wellposedness $\Leftrightarrow$ $\underbrace{\begin{bmatrix} I & -\hat{K} \\ -P & I \end{bmatrix}}_{\text{Transfer matrix}}$ is invertible and proper

$\leftarrow \hat{K}(s)$

Rational polynomial matrix

$$\Updownarrow$$

$$\begin{bmatrix} I & -\hat{K}(\infty) \\ -P(\infty) & I \end{bmatrix} \text{ is invertible} \quad \# \quad \begin{bmatrix} I & -P(\infty) \\ -\hat{K}(\infty) & I \end{bmatrix} \text{ is invertible}$$

$\Updownarrow$

$I - K(\infty)P(\infty)$ is invertible.

$$* \quad \begin{bmatrix} I & -\hat{K}(\infty) \\ -P(\infty) & I \end{bmatrix} \text{ is invertible} \iff I - P(\infty)\hat{K}(\infty) \text{ is invertible}$$

$$\checkmark \quad \underbrace{\begin{bmatrix} I & -\hat{K}(\infty) \\ -P(\infty) & I \end{bmatrix}}_{\text{if this is invertible,}} = \underbrace{\begin{bmatrix} I & 0 \\ -P(\infty) & \boxed{I - P(\infty)\hat{K}(\infty)} \end{bmatrix}}_{\text{then needs to be invertible}} \underbrace{\begin{bmatrix} I & -K(\infty) \\ 0 & I \end{bmatrix}}_{\because \text{ this is invertible}} \checkmark$$

$$P(s) := \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right], \quad \hat{K}(s) := \left[\begin{array}{c|c} \hat{A} & \hat{B} \\ \hline \hat{C} & \hat{D} \end{array}\right]$$

Well-posed: $P(\infty) = D$ $\qquad$ $\hat{K}(\infty) = \hat{D}$

System is well-posed $\iff$ $\begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix}$ is invertible

$\iff$ $(I - \hat{D}D)$ is invertible

$\iff$ $(I - D\hat{D})$ is invertible

$\iff$ $\begin{bmatrix} I & -D \\ -\hat{D} & I \end{bmatrix}$ is invertible

Practical systems: $D = 0$ (Plant doesnot generally have a feedthrough term).

$\begin{bmatrix} I & -\hat{D} \\ 0 & I \end{bmatrix} \leftarrow$